

HW50 Kafka

Explain following concepts, and how they coordinate with each other:

- Topic
- Partition
- Broker
- Consumer group
- Producer
- Offset
- Zookeeper

Answer following questions:

1. Given N (number of partitions) and M (number of consumers,) what will happen when $N \geq M$ and $N < M$ respectively?
2. Explain how brokers work with topics?
3. Are messages pushed to consumers or consumers pull messages from topics?
4. How to avoid duplicate consumption of messages?
5. What will happen if some consumers are down in a consumer group? Will data loss occur? Why?
6. What will happen if an entire consumer group is down? Will data loss occur? Why?
7. Explain consumer lag and how to resolve it?
8. Explain how Kafka tracks message delivery?
9. Compare Kafka vs RabbitMQ, compare messageing frameworks vs MySQL (Why Kafka)?
10. On top of <https://github.com/CTYue/Spring-Producer-Consumer>
 - Write your consumer application with Spring Kafka dependency, set up 3 consumers in a single consumer group.
Prove message consumption with screenshots.
 - Increase number of consumers in a single consumer group, observe what happens, explain your observation.
 - Create multiple consumer groups using Spring Kafka, set up different numbers of consumers within each group, observe consumer offset,
 - Prove that each consumer group is consuming messages on topics as expected, take screenshots of offset records,

- Demo different message delivery guarantees in Kafka, with necessary code or configuration changes.
- Write your own partitioning logic, to demo your messages will be assigned to different brokers, or you can also demo kafka's default round-robin partitioning logic, with code changes.

Note: There's already a docker compose file in above git repo, you can start the 3 brokers (with zookeep) using docker-compose up command.